Manuscript title: Semantic-Guided Fusion Network for Multi-source Remote Sensing Image Classification

Article ID: GRSL-01407-2026

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Summary of Changes

Dear Editors,

We sincerely thank you for your time and invaluable feedback during the review process. Your expert comments have been instrumental in improving the quality and clarity of our manuscript. We have carefully addressed every point raised and made comprehensive revisions accordingly. All modifications in the revised manuscript are highlighted in red for easy reference. We have also added more detailed explanations and analyses to ensure that the manuscript is more comprehensive and informative.

We hope these revisions meet your expectations. Should further clarifications or adjustments be needed, we are fully committed to addressing them promptly. Your feedback is highly valued, and we are grateful for your support in helping us improve our work.

Sincerely,

Yuwei Zhao and Feng Gao

Ocean University of China

Email: gaofeng@ouc.edu.cn

Response to the Associate Editor

Overall Comment

This letter present a semantic-guided fusion network for multi-source remote sensing image classification. However, several critical issues need to be addressed.

Response:

We greatly appreciate the Associate Editor and the reviewers for their constructive comments and insightful recommendations to improve the quality of the manuscript. We have revised the manuscript extensively, addressing all the comments provided. Below is a summary of the major changes made to the paper, followed by detailed responses to each reviewer's comment.

The major revisions of the paper are as follows:

AAA:

aaa

BBB:

bbb

CCC:

ccc

DDD:

ddd

EEE:

eee

FFF:

fff

GGG:

ggg

HHH:

hhh

In the following, we will respond to each of the comments in detail.

Response to Reviewer 1

We would like to thank you for your valuable comments and suggestions. More importantly, we are very pleased that we can have so many beneficial discussions on multisource remote sensing image classification with you. We believe these discussions must be (and it has been) helpful for us to improve this work.

We have very carefully followed all the comments and cleared up every issue raised. We provide below a point-by-point response to the comments as follows.

Reviewer #1, Comment #1

The manuscript does not clearly explain how the proposed SMCB differs substantially from existing dynamic convolution or semantic-aware filtering methods. Please provide in-depth description to highlight the technical contribution of the proposed method.

Response:

TODO

Reviewer #1, Comment #2

Please explain why frequency-domain modulation is more robust to misalignment and provide controlled experiments with different degrees of artificial spatial shifts to verify this claim.

Response:

TODO

Reviewer #1, Comment #3

Eq. 11 and 12 appear to place the value features inside the Softmax operation, which is inconsistent with the standard attention formulation. Please check carefully or provide more descriptions.

Response:

TODO

Reviewer #1, Comment #4

The paper states that the query and key features are multiplied to obtain an affinity map, and then a dynamic kernel is generated. However, the exact dimensions of the affinity map, the reshaping operation, and the dynamic kernel are not sufficiently clear. Please add more discussions.

Response:

TODO

Reviewer #1, Comment #5

The comparison should be expanded to include more recent and stronger baselines.

Response:

TODO

Reviewer #1, Comment #6

Although SGFNet achieves the best overall accuracy, it does not consistently outperform all baselines for every class. This indicates that the proposed method may still struggle with some spectrally ambiguous or minority classes. The authors should provide a deeper analysis of these failure cases.

Response:

TODO

Thanking you again for your precious time and valuable efforts spent in reviewing our manuscript. We greatly appreciate your expertise and the dedicated efforts you have made to assist us in enhancing the quality of this manuscript. We hope that we have clarified all the issues raised by you, and our revised manuscript will again meet your expectations. We eagerly anticipate any additional insights or suggestions you might offer.

Best wishes and thanks again.

Yuwei Zhao and Feng Gao

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Response to Reviewer 2

We would like to thank you for your valuable comments and suggestions. More importantly, we are very pleased that we can have so many beneficial discussions on multisource remote sensing image classification with you. We believe these discussions must be (and it has been) helpful for us to improve this work.

We have very carefully followed all the comments and cleared up every issue raised. We provide below a point-by-point response to the comments as follows.

Reviewer #2, Comment #1

Has the author considered the issue of class imbalance? From the experimental results, such a problem indeed exists. What are the underlying causes of this issue?

Response:

TODO

Reviewer #2, Comment #2

How does SGFNet address the issue of spatial misalignment in multi-source data?

Response:

TODO

Reviewer #2, Comment #3

The paper mentions that SMCB can better capture long-range semantic dependencies compared to traditional convolution, and how this can be explained through physical meaning.

Response:

TODO

Reviewer #2, Comment #4

What impact do different convolution kernel sizes in the SMCB module have on the overall detection performance?

Response:

TODO

Thanking you again for your precious time and valuable efforts spent in reviewing our manuscript. We greatly appreciate your expertise and the dedicated efforts you have made to assist us in enhancing the quality of this manuscript. We hope that we have clarified all the issues raised by you, and our revised manuscript will again meet your expectations. We eagerly anticipate any additional insights or suggestions you might offer.

Best wishes and thanks again.

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Response to Reviewer 3

We would like to thank you for your valuable comments and suggestions. More importantly, we are very pleased that we can have so many beneficial discussions on multisource remote sensing image classification with you. We believe these discussions must be (and it has been) helpful for us to improve this work.

We have very carefully followed all the comments and cleared up every issue raised. We provide below a point-by-point response to the comments as follows.

Reviewer #3, Comment #1

SMCB involves dynamic kernel generation and semantic affinity computation, while FMFB introduces attention-like operations. These operations may increase computational cost. Please add corresponding efficiency analysis.

Response:

TODO

Reviewer #3, Comment #2

Some parameters, such as the dynamic kernel size in SMCB, the channel dimension is not analyzed or introduced. Please add more sensitivity analysis or descriptions.

Response:

TODO

Reviewer #3, Comment #3

Although SGFNet improves the overall performance, the classification accuracy of difficult categories remains unsatisfactory. The authors should discuss why these classes are still poorly classified.

Response:

TODO

Reviewer #3, Comment #4

The classification maps of Augsburg are not provided. Please add more visualized classification results.

Response:

TODO

Reviewer #3, Comment #5

The relationship between DCT-domain modulation and spatial misregistration robustness is only qualitatively described. Please provide more evidence or discussions.

Response:

TODO

Reviewer #3, Comment #6

In Fig. 3, the role of the frequency-domain difference feature should be further explained. It is unclear whether the difference operation may suppress useful modality-specific information.

Response:

TODO

Thanking you again for your precious time and valuable efforts spent in reviewing our manuscript. We greatly appreciate your expertise and the dedicated efforts you have made to assist us in enhancing the quality of this manuscript. We hope that we have clarified all the issues raised by you, and our revised manuscript will again meet your expectations. We eagerly anticipate any additional insights or suggestions you might offer.

Best wishes and thanks again.

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